SVKM'S NMIMS

MUKESH PATEL SCHOOL OF TECHNOLOGY MANAGEMENT ENGINEERING / SCHOOL OF TECHNOLOGY MANAGEMENT ENGINEERING

Program: B. Tech (CSBS) Year: II Semester : III

AcademicYear: 2022-2023

Subject: Object Oriented Programming

Marks: 100

Date : 30 January 2023

Time: 10.00 am - 1.00 pm

Durations: 3 (Hrs)

No. of Pages: 03.

Re-Examination (2022-23)

Instructions: Candidates should read carefully the instructions printed on the question paper and on the cover of the Answer Book, which is provided for their use.

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1) Question No. 1 is compulsory.
2) Out of remaining questions, attempt any 4 questions.
3) **In all 5 questions to be attempted.**
4) All questions carry equal marks.
5) **Answer to each new question to be started on a fresh page.**
6) **Figures in brackets on the right hand side indicate full marks.**
7) **Assume Suitable data if necessary.**

Q1		Answer briefly:	[20]
CO- 2 BL-2	a.	Explain one dimensional array. How insertion and deletion are done in same.	[05]
CO- 2 BL-2	b.	Explain the usage of scope resolution operators with respect to classes and objects.	[05]
CO-2 BL-1,2	c.	List the advantages of using inheritance. Write syntax demonstrating Single and multiple inheritance.	[05]
CO- 4 BL-2	d.	Explain extends and includes with respect to use case diagram	[05]
Q2 CO-1 BL-2	a	Define procedural and object oriented programming? Elaborate macros with the help of an example..	[10]
CO-3; BL-3	b	Which header file needed to write in a file? Illustrate using syntax. Write a program to read from a file.	[10]
Q3 CO-1 BL-3	a	What is the role of inline function in object oriented programming language? Demonstrate with the help of a program to add, subtract, multiply and divide two numbers.	[10]
CO-3 BL-2,3	b	What is the need of template class? Demonstrate with the help of an example	[10]

Q4 CO-2 BL-3	a	Discuss the use of Access specifiers in C++. Write down a program showing the use of all types of access specifiers clearly.	[10]
CO-4 BL-6	b	Design Use case diagram for library Management system. Write down various notations	[10]
Q5 CO-1 BL-3	a	Compare pass by value and pass by reference. Illustrate with the help of a program.	[10]
CO-4 BL-6	b	Design class Diagram for Airline Reservation System	[10]
Q6 CO-2 BL-6	a	Create a class Hospital having data members as name, address and number of branches accept the data in member function acceptinfo, Create class Doctor inheriting Hospital with data member as name, age , specialization and Hospital name. accept the details in the function getdocinfo. After accepting the hospital name, check whether the hospital name is the same as the Hospital class if yes then display the name of hospital, docname, specialization and Fess. If the hospital name is different, then display Doctor is not tied up with the hospital.	[10]
CO-2 BL-6	b	Write a program to calculate the results of a student by combining internal and external marks out of 100. Write two separate classes as "Internal" and "External1" with following data members common to both the classes. (out of 50 marks for each class) · Marks of Math · Marks of OPP · Marks of SE · Marks of DS Use a constructor to input the marks for each class. Define binary operator overloading **using friend function** to calculate the final result by combining internal and external marks. Perform following tasks: 1. Print the total marks subject wise. 2. Print the result depending on marks of individual subjects (Marks < 50 - fail) 3. Calculate total marks 4. Print percentage if student pass	[10]
Q7 CO-2 BL-3	a	Differentiate function overloading and overriding. Write syntax depicting overloading and overriding.	[05]
CO-2 BL-2	B	Explain Data abstraction with the help of an example	[05]

CO2, BL2	C	Is there any universal catch available to handle the exception? If yes, then demonstrate it using a program.	[05]
CO-2 BL-2	d	Explain the concept of destructor in C++ with the help of an example.	[05]